

Gibbs energy and spontaneous changes (HL)

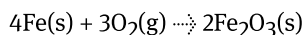
Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

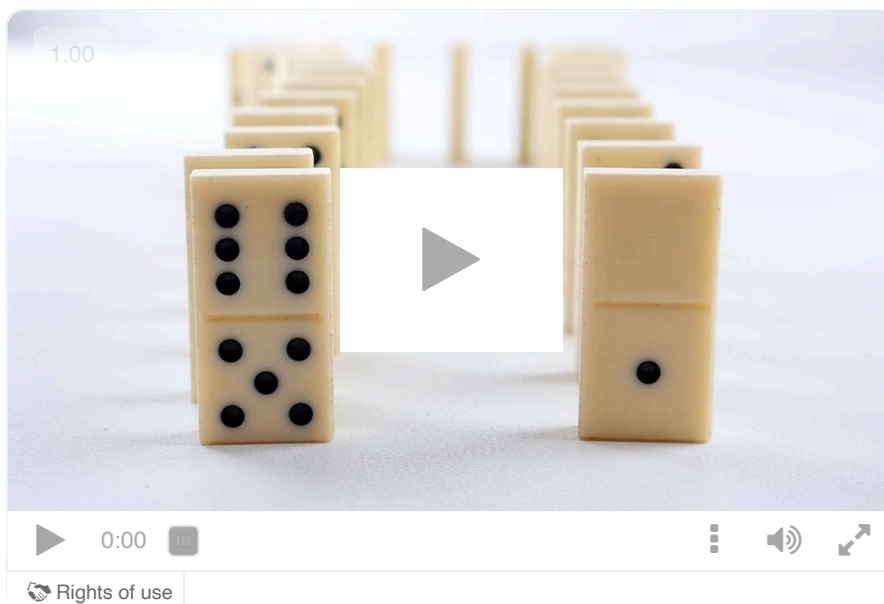
- (R1.4.2) Apply the equation $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$ to calculate unknown values of these terms.
- (R1.4.3) Interpret the sign of ΔG calculated from thermodynamic data.
- (R1.4.3) Determine the temperature at which a reaction becomes spontaneous.

In our daily lives, we encounter many reactions that happen without the need of consistent energy input. For example, when you strike a match to paper, the paper will continue burning even after you blow the match. If an iron nail is exposed to air rusts (iron(III) oxide) forms, according to the equation:



However, iron(III) oxide (rust) will not break down to form iron and oxygen, nor can you return the paper to its original state once it has burned. The burning of paper and the rusting of a nail are known as spontaneous reactions. Their inverse reactions are called non-spontaneous reactions.

Spontaneous reactions will happen once the energy needed to start the reaction is provided. Like the touch that the finger gives to the first domino in **Video 1**:



Video 1. Visualisation of the Activation Energy Needed to Start a Spontaneous Reaction.

Credit: magicmore, Getty Images

(<https://www.gettyimages.com/detail/video/white-dominoes-falling-in-chain-reaction-domino-effect-stock-footage/853285342>)

 More information for video 1

An interactive video features a series of vertically arranged dominoes forming an almost inverted U-shape. As the video progresses, a person's finger gradually pushes the first domino on the left, which has eleven dots, causing it to fall and knock over the next domino. The dominoes keep toppling one by one until the last one on the right, with one dot, falls.

This falling domino effect represents a spontaneous reaction, where an initial push sets off a continuous process without requiring additional input.

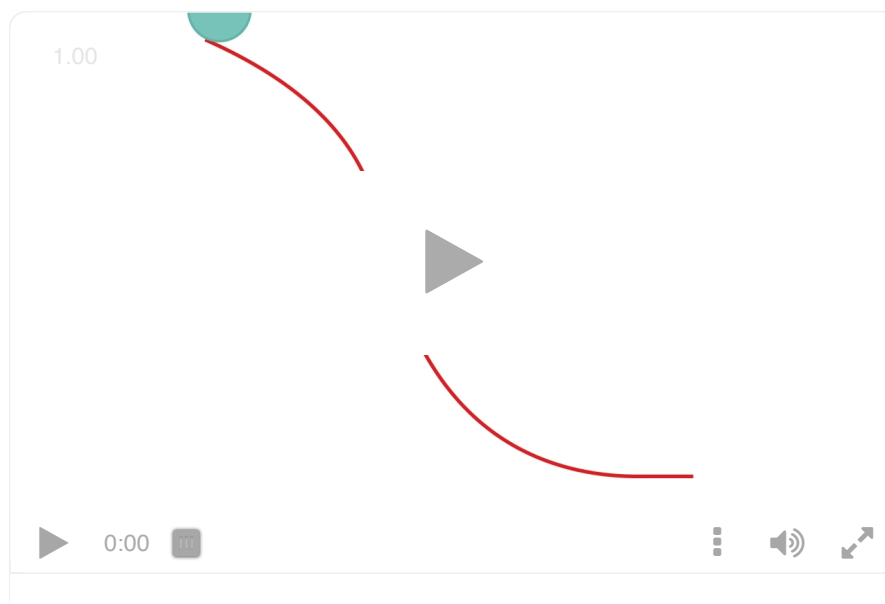
How can we determine if a reaction will happen spontaneously?

What factors determine the feasibility of a reaction?

Spontaneous reactions

A spontaneous reaction occurs without the addition of energy, other than the energy required to start the reaction (also called activation energy). For example, if you strike a match to paper, the paper will continue to burn by itself, even if you remove the match – this is a spontaneous reaction. But if you want to fry potato chips, you need to continuously provide heat until they are cooked – this is a non-spontaneous reaction.

A spontaneous process can be compared to a ball rolling down a hill; once the ball is given the initial push, it will continue to roll down the hill until it reaches the bottom (**Interactive 1**). A non-spontaneous reaction is one that needs a constant input of energy to occur. It can be compared to a ball rolling up a hill, which only occurs if energy is constantly supplied to get the ball to the top.



Interactive 1. A Spontaneous and a Non-spontaneous Process.

 More information for interactive 1

The interactive video is shown with a blue ball placed at the top of a curved red hill. The interactive compares spontaneous and non-spontaneous processes.

Spontaneous Process: A ball rolls down a downward-sloping curve smoothly, representing a process that occurs naturally without external input.

Non-spontaneous Process: The same ball moves uphill along the same curve, indicating that an external force or energy is required for the process to occur.

This interactive helps users understand the difference between spontaneous and non-spontaneous processes by illustrating energy changes through ball movement.

Making connections

Activation energy is the energy required to start a chemical reaction. You will learn more about it in [section R2.2.4](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>).

The second law of thermodynamics (**Figure 1**) states that the total entropy of the universe tends to increase. Simply put, in a spontaneous process, the energy goes from a concentrated form to a dispersed form (entropy increases). For a spontaneous process, the total entropy of the system and surroundings (ΔS_{total}) must increase. This can be represented in equation form as:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

Certain chemical reactions have a decrease in the entropy of the system but are still spontaneous because the entropy of the surroundings increases to a much greater extent, which gives an overall increase in entropy for the process. One example is the formation of crystals in a salt solution when the solvent evaporates. The crystals (system) have a lower entropy (more ordered state) than the salt solution, but the solvent molecules increase in entropy as they are lost to the surrounding as a gas (more disordered state).

The same happens with our planet as part of the universe. It was mentioned in the last section that the universe is expanding, which means it is becoming more disordered – its entropy is increasing. However, on Earth we can find many examples of ordered systems, like the rocks that make up most of the planet. More order means that the entropy of the system is lower. Nonetheless, we need to consider that overall entropy is still increasing, as the surroundings of our planet (the universe) are increasing in entropy faster than entropy decreases in our planet (the system).

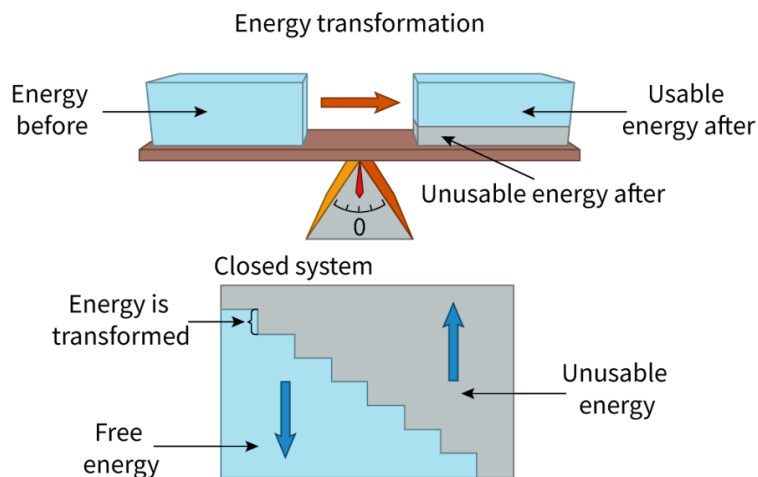


Figure 1. The second law of thermodynamics.

 More information for figure 1

The diagram illustrates concepts related to thermodynamics. At the top, there is a seesaw-like setup with two rectangular blocks of ice placed on it, moving left and right. Arrows indicate the directions of movement, with an orange arrow showing the general direction from left to right. At the fulcrum, there is a triangular indicator set at zero, suggesting equilibrium. Below, there is a gradient, step-like schematic with arrows. A small arrow pointing to the left is shown on the left side, and larger arrows show movement downward and upward. Components are labeled with descriptive text explaining different parts of the process, potentially relating to heat exchange, equilibrium, and energy transformation processes in thermodynamics.

[Generated by AI]

Gibbs energy (G)

The increase in the entropy of the surroundings is usually in the form of heat released from the system into the surroundings. You can feel it if you place your hand close to a fire.

For the surroundings, the entropy change is given by the relationship in **Table 1**.

Table 1. Equation for entropy change in the surrc

Equation	Symbols
$\Delta S_{\text{surroundings}} = -\Delta H/T$	$\Delta S_{\text{surroundings}}$ = the entropy change
	ΔH = the enthalpy change of the process
	T represents the absolute temperature

Notice the change of sign: the heat given out by the system (ΔH for the system will be negative) enters the surroundings, so any decrease in the enthalpy of the system equates to an addition of heat to the surroundings ($\Delta S_{\text{surroundings}}$ will be positive). This equation can be transformed to the one in **Table 2**.

Table 2. Equation for the change in standard Gibbs energy under standard conditions temperature and pressure.

Equation	Symbols	Unit
$\Delta G^{\ominus} = \Delta H^{\ominus} - T\Delta S^{\ominus}$	$\Delta G^{\ominus}$ = the change in standard Gibbs energy under standard conditions.	kilojoule mol ⁻¹ (kJ mol ⁻¹)
	$\Delta H^{\ominus}$ = the standard enthalpy under standard conditions.	kilojoule mol ⁻¹ (kJ mol ⁻¹)
	T = temperature	kelvin
	$\Delta S^{\ominus}$ = the standard entropy change	joules per mole per kelvin (J K ⁻¹ mol ⁻¹)

A change in Gibbs energy, ΔG , takes into account the direct entropy change resulting from the transformation of the chemicals and the indirect entropy change of the surroundings resulting from the transfer of heat energy.

Study skills

Thermodynamic data values are given in the DP chemistry data booklet in **section 13** (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>).

When calculating ΔG , carefully check the units of ΔH (which are generally kJ mol⁻¹) and ΔS (which are generally J K⁻¹ mol⁻¹) and convert them if necessary. All temperatures must be in kelvin.

Generally, you will want to convert joules to kilojoules, which means you need to divide the number by 1000.

Nature of Science

Aspect: Models

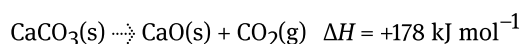
For DP chemistry, in the context of studying entropy, models can be used to predict whether the entropy change for a given reaction or process is positive or negative. For example, the calculation of the standard entropy change of a reaction uses a model that considers the absolute entropy

values of each substance involved in the reaction and the number of moles of each substance to calculate the total absolute entropy on each side of the reaction. The difference between the total absolute entropy on the left-hand side and the right-hand side gives us the change in entropy for the reaction. This model allows us to predict whether the entropy change for a given reaction or process is positive or negative.

It is now clear that there are three factors to be considered when determining the spontaneity of a reaction:

- the sign of ΔH
- the sign of ΔS
- the temperature at which the reaction takes place.

This allows us to explain the occurrence of endothermic reactions, such as the decomposition of calcium carbonate to form calcium oxide and carbon dioxide:



Work through these two examples to learn why endothermic reactions can be spontaneous.

Worked example 1

Calculate the ΔG for the decomposition of calcium carbonate at 298 K knowing that ΔH for the reaction is $+178 \text{ kJ mol}^{-1}$ and the ΔS is $+161 \text{ J K}^{-1} \text{ mol}^{-1}$.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$\Delta H = +178 \text{ kJ mol}^{-1}$ $\Delta S = +161 \text{ J K}^{-1} \text{ mol}^{-1}$ $T = 298 \text{ K}$
Step 2: Convert any values that need changing.	ΔS need dividing by 1000 $\Delta S = \frac{+161}{1000}$ $= +0.161 \text{ kJ K}^{-1} \text{ mol}^{-1}$
Step 3: Write out the equation.	$\Delta G = \Delta H - T\Delta S$
Step 4: Substitute the values given.	$= +178 - 298 \times +0.161$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= +130 \text{ kJ mol}^{-1}$

Worked example 2

Calculate the ΔG for the decomposition of calcium carbonate at 1106 K knowing that ΔH for the reaction is $+178 \text{ kJ mol}^{-1}$ and the ΔS is $+161 \text{ J K}^{-1} \text{ mol}^{-1}$.

Solution steps	Calculati
Step 1: Write out the values given in the question.	$\Delta H = +$ $\Delta S = +$ $T = 1106$
Step 2: Convert any values that need changing.	ΔS need dividing by $\Delta S = \frac{+}{1}$ $= +$
Step 3: Write out the equation.	$\Delta G = \Delta$
Step 4: Substitute the values given.	$= (+178$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= -0.061$

As you can see from these calculations, Gibbs energy change (ΔG) can be either positive or negative. For any reaction, the sign of ΔG is dependent only on the temperature at which the reaction is happening, as the enthalpy and entropy are the same for the same reaction. A reaction is spontaneous when ΔG is negative.

Fill in the blank in **Interactive 2** to complete the statement.

Fill in the missing number

So for the decomposition of calcium carbonate, the reaction is spontaneous when the temperature is K or above.

☒ Check

Interactive 2. Fill in the Blank.

The spontaneity of a reaction is dependent on the sign of the Gibbs energy change (ΔG). This is the ultimate deciding factor that tells us whether a reaction is spontaneous or not, as follows:

- If the Gibbs energy change, ΔG , is negative, the reaction is spontaneous.
- If the Gibbs energy change, ΔG , is positive, the reaction is non-spontaneous.
- If the Gibbs energy change, ΔG , is zero, the reaction is at equilibrium.

Using the equation $\Delta G = \Delta H - T\Delta S$, complete **Interactive 3**, which summarises the factors affecting the spontaneity of a reaction.

	$\Delta H < 0$	$\Delta H > 0$
$\Delta S > 0$		
$\Delta S < 0$		

Non-spontaneous at all T ($\Delta G > 0$)	Spontaneous at low T (when $T\Delta S$ is small)	Spontaneous at all T ($\Delta G < 0$)	Spontaneous at high T (when $T\Delta S$ is large)
---	--	---	---

✓ Check

Interactive 3. Complete the table.

 More information for interactive 3

An interactive table with drop boxes represents the factors affecting the spontaneity of a reaction. The table contains two columns labeled $\Delta H < 0$ and $\Delta H > 0$, and two rows labeled $\Delta S > 0$ and $\Delta S < 0$. Each row or column has two dotted drop boxes.

Users need to drag and drop the descriptions of spontaneity into the appropriate drop boxes in the table.

The boxed drag-and-drop options provided at the bottom from left to right are:

Non-spontaneous at all T ($\Delta G > 0$)

Spontaneous at low T (when $T\Delta S$ is small)

Spontaneous at all T ($\Delta G < 0$)

Spontaneous at high T (when $T\Delta S$ is large)

This interactivity helps users understand how enthalpy (ΔH) and entropy (ΔS) influence the Gibbs free energy (ΔG) and the spontaneity of chemical

reactions.

The “Check” icon at the bottom left allows users to check their answers. The right answer is indicated with a slider. If the inputted answers are wrong, the “Retry” icon at the bottom left allows users to retry the activity.

Read below for the solution.

Column 1 Row 1: Spontaneous at all T ($\Delta G < 0$)

Column 2 Row 1: Spontaneous at high T (when $T\Delta S$ is large)

Column 1 Row 2: Spontaneous at low T (when $T\Delta S$ is small)

Column 2 Row 2: Non-spontaneous at all T ($\Delta G > 0$)

Looking at **Interactive 3**, you can see that there are two cases in which the value of the ΔG is independent of the temperature:

- Exothermic reactions that result in an increase in entropy — the value of the ΔG will be negative at any temperature:

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (-) - (+)(+) \\ &= (-) \quad \text{always spontaneous}\end{aligned}$$

- Endothermic reactions with a decrease in entropy — the value of the ΔG is positive at any temperature.

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (+) - (+)(-) \\ &= (+) \quad \text{never spontaneous}\end{aligned}$$

Temperature is a factor in the other two scenarios:

- Exothermic reactions with a decrease in entropy are only spontaneous at low temperatures.

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (-) - (+)(-) \\ &= (-) \text{ if } T \text{ is low (spontaneous) OR} \\ &\quad (+) \text{ if } T \text{ is high (non-spontaneous)}\end{aligned}$$

- Endothermic reactions with an increase in entropy are only spontaneous at high temperatures. If the temperature is high enough, subtracting the product of $T\Delta S$ from the ΔH value will give a negative (spontaneous) ΔG value. At lower temperatures, the product of $T\Delta S$ is not large enough to cancel out the positive ΔH value, so the ΔG value is positive (non-spontaneous).

$$\begin{aligned}\Delta G &= \Delta H - T\Delta S \\ &= (+) - (+)(+) \\ &= (-) \text{ if } T \text{ is high (spontaneous) OR} \\ &\quad (+) \text{ if } T \text{ is low (non-spontaneous)}\end{aligned}$$

Making connections

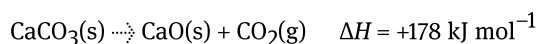
Section R1.3.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>) — Cells need to have a spontaneous reaction to provide electricity. What does it mean with respect to Gibbs energy?

You might notice that batteries you buy from the store are labelled with the voltage (1.5 V, 6 V, 9 V, etc). This measurement of the energy a cell can provide is known as the cell potential. The cell potential is a positive value and you have learned in this section that for the reaction to be spontaneous, Gibbs energy must be a negative value. What is the relationship between the cell potential and the Gibbs energy for an electrochemical cell?

Making connections

In section R3.2.14 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/14%20gibbs-free-energy-and-standard-electrode-potentials-hl-id-44784/>), you will learn more about cells and calculating their potential. How can electrochemical data also be used to predict the spontaneity of a reaction?

You have already seen the equation and enthalpy for the decomposition of calcium carbonate:



Apply your knowledge to this reaction to answer the following questions.

Worked example 3

State whether the decomposition of calcium carbonate is endothermic or exothermic.

Endothermic, as the enthalpy change is positive.

Worked example 4

What can you deduce about the entropy for the reaction?

Entropy is positive as there is more disorder in the products than reactants due to one mole of solid being converted into one mole of solid and one mole of gas.

Worked example 5

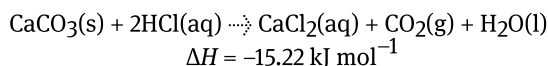
What temperature conditions are likely to lead to a spontaneous reaction. Justify your answer.

$$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$$

The reaction will become spontaneous at elevated temperatures as $-T\Delta S^\ominus$ will become greater than $\Delta H^\ominus$ which will lead to a negative $\Delta G^\ominus$.

At lower temperatures, $\Delta G^\ominus \approx \Delta H^\ominus$, which means Gibbs energy will be positive and the reaction will be non-spontaneous.

Consider the reaction of calcium carbonate with hydrochloric acid and answer the following questions.



Worked example 6

Predict, stating a reason, the sign of $\Delta S^\ominus$ for the above reaction.

The sign of $\Delta S^\ominus$ is positive because one mole of solid plus two moles of aqueous reactants produce one mole of aqueous plus one mole of gaseous plus one mole of liquid products, so there is an increase in entropy.

Worked example 7

Calculate the standard entropy change for the reaction.

Note:

$$S^\ominus (\text{CaCO}_3(\text{s})) = +38.1 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{HCl}(\text{aq})) = +53.1 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S^\ominus (\text{CaCl}_2(\text{aq})) = +124 \text{ J K}^{-1} \text{ mol}^{-1}$$

Use section 13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet.

Solution steps	Calculations
Step 1: Write out the values given in the question and obtain any other values you need from the DP chemistry data booklet.	$S^\ominus (\text{CaCO}_3(\text{s})) = +38.1 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{HCl}(\text{aq})) = +53.1 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{CaCl}_2(\text{aq})) = +124 \text{ J K}^{-1} \text{ mol}^{-1}$ From the DP chemistry data booklet $S^\ominus (\text{CO}_2(\text{g})) = +214 \text{ J K}^{-1} \text{ mol}^{-1}$ $S^\ominus (\text{H}_2\text{O}(\text{l})) = +70 \text{ J K}^{-1} \text{ mol}^{-1}$
Step 2: Write out the equation.	$\Delta S^\ominus = \Delta S^\ominus (\text{products}) - \Delta S^\ominus (\text{reactants})$
Step 3: Substitute the values given.	$= [(+124) + (+214) + (+70)] - [$

Solution steps	Calculations
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$\Delta S^{\ominus} = +264 \text{ J K}^{-1} \text{ mol}^{-1}$ (3 s.f.)

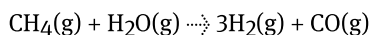
Worked example 8

Determine the value of $\Delta G^{\ominus}$ for the reaction at 298 K and explain if the reaction is spontaneous at this temperature.

Solution steps	Calculations
Step 1: Use the values previously presented /calculated.	$\Delta H^{\ominus} = -15.22 \text{ kJ mol}^{-1}$ $\Delta S^{\ominus} = +264 \text{ J K}^{-1} \text{ mol}^{-1}$
Step 2: Convert any values that need changing.	$\Delta S^{\ominus} = +0.264 \text{ kJ K}^{-1} \text{ mol}^{-1}$
Step 3: Write out the equation.	$\Delta G^{\ominus} = \Delta H^{\ominus} - T \Delta S^{\ominus}$
Step 4: Substitute the values given.	$= (-15.22) - (298)(0.264)$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= -93.9 \text{ kJ (3 s.f.)}$ Given that $\Delta G^{\ominus}$ is negative, the reaction is spontaneous at this temperature.

Worked example 9

The reaction between methane and steam is non-spontaneous at 298 K.



Determine the temperature above which the reaction becomes spontaneous given the following data:

$$\Delta H = 205 \text{ kJ mol}^{-1}$$

$$\Delta S = 216 \text{ J K}^{-1} \text{ mol}^{-1}$$

Solution steps	Calculations
Step 1: Use the information provided in the question.	For the reaction to be spontaneous you calculate the temperature at which $\Delta G^\ominus = 0$ to tell at which temperature the reaction becomes spontaneous.
Step 2: Convert any values that need changing.	$\Delta S^\ominus = 0.216 \text{ kJ K}^{-1} \text{ mol}^{-1}$
Step 3: Write out the equation.	$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$
Step 4: Substitute the values given.	$0 = (205) - T(0.216)$ $-205 = -0.216T$ $T = \frac{-205}{-0.216}$ $= 949$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 949 \text{ K (3 s.f.)}$ This reaction becomes spontaneous above 949 K (or 1222 °C).

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Consider the boiling of water.

- Write a balanced chemical equation to represent the change of state of water.
- Use section 13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of your DP chemistry data booklet to calculate the enthalpy of this process.

Explain why the boiling point of pure water is 373 K.

- Use your knowledge of intermolecular forces to choose another substance that has a lower boiling point than water. Show calculations to support your choice.